

EXTENDS *Naturals*

\*\*\*\*\*

This model keeps only the states needed to visualize three delayed operations: a transfer confirmation, an old *G1* page logout, and old-cookie cleanup. *G3A* represents any successor installed while one of those operations is delayed.

*RunOldPageCleanup* treats inspection and mutation as one action. This is a protocol model under the implementation's enforced browser-profile Web Lock, which spans command creation, both *HTTP* responses, and response-side cleanup. It does not claim that the cleanup would be safe without that serialization.

\*\*\*\*\*

$Sessions \triangleq \{ \text{"G1A"}, \text{"G2B"}, \text{"G3A"} \}$   
 $Cookies \triangleq \{ \text{"G1"}, \text{"G2"}, \text{"G3"}, \text{"none"} \}$   
 $ChallengeStates \triangleq \{ \text{"none"}, \text{"pending"}, \text{"used"} \}$

VARIABLES

*currentSession*,  
*epoch*,  
*cookieA*,  
*cookieB*,  
*challengeState*,  
*successorInstalled*,  
*revoked*,  
*oldLogoutHandled*,  
*oldCleanupHandled*

$vars \triangleq \langle currentSession, epoch, cookieA, cookieB, challengeState, \\ successorInstalled, revoked, oldLogoutHandled, \\ oldCleanupHandled \rangle$

*Init*  $\triangleq$

$\wedge currentSession = \text{"G1A"}$   
 $\wedge epoch = 1$   
 $\wedge cookieA = \text{"G1"}$   
 $\wedge cookieB = \text{"none"}$   
 $\wedge challengeState = \text{"none"}$   
 $\wedge successorInstalled = \text{FALSE}$   
 $\wedge revoked = \text{FALSE}$   
 $\wedge oldLogoutHandled = \text{FALSE}$   
 $\wedge oldCleanupHandled = \text{FALSE}$

*StageTransfer*  $\triangleq$

$\wedge challengeState = \text{"none"}$   
 $\wedge challengeState' = \text{"pending"}$   
 $\wedge \text{UNCHANGED } \langle currentSession, epoch, cookieA, cookieB, \\ successorInstalled, revoked, oldLogoutHandled, \\ oldCleanupHandled \rangle$

*oldCleanupHandled*⟩

*ConfirmTransfer*  $\triangleq$   
 $\wedge$  *challengeState* = “pending”  
 $\wedge$  *challengeState'* = “used”  
 $\wedge$  IF *currentSession* = “G1A”  $\wedge$  *epoch* = 1  $\wedge$   $\neg$ *revoked*  
    THEN  $\wedge$  *currentSession'* = “G2B”  
     $\wedge$  *epoch'* = 2  
     $\wedge$  *cookieB'* = “G2”  
     $\wedge$  UNCHANGED ⟨*cookieA*, *revoked*⟩  
    ELSE  $\wedge$  UNCHANGED ⟨*currentSession*, *epoch*, *cookieA*, *cookieB*,  
    *revoked*⟩  
 $\wedge$  UNCHANGED ⟨*successorInstalled*, *oldLogoutHandled*,  
*oldCleanupHandled*⟩

*InstallCompetingSuccessor*  $\triangleq$   
 $\wedge$  *challengeState*  $\neq$  “none”  
 $\wedge$   $\neg$ *revoked*  
 $\wedge$  *currentSession*  $\neq$  “G3A”  
 $\wedge$  *currentSession'* = “G3A”  
 $\wedge$  *epoch'* = *epoch* + 1  
 $\wedge$  *cookieA'* = “G3”  
 $\wedge$  *successorInstalled'* = TRUE  
 $\wedge$  UNCHANGED ⟨*cookieB*, *challengeState*, *revoked*,  
*oldLogoutHandled*, *oldCleanupHandled*⟩

*RunOldPageLogout*  $\triangleq$   
 $\wedge$  *currentSession*  $\neq$  “G1A”  
 $\wedge$   $\neg$ *oldLogoutHandled*  
 $\wedge$  *oldLogoutHandled'* = TRUE  
 $\wedge$  *revoked'* = IF *currentSession* = “G1A” THEN TRUE ELSE *revoked*  
 $\wedge$  UNCHANGED ⟨*currentSession*, *epoch*, *cookieA*, *cookieB*,  
*challengeState*, *successorInstalled*,  
*oldCleanupHandled*⟩

*RunOldPageCleanup*  $\triangleq$   
 $\wedge$  *currentSession*  $\neq$  “G1A”  
 $\wedge$   $\neg$ *oldCleanupHandled*  
 $\wedge$  *oldCleanupHandled'* = TRUE  
 $\wedge$  *cookieA'* = IF *cookieA* = “G1” THEN “none” ELSE *cookieA*  
 $\wedge$  UNCHANGED ⟨*currentSession*, *epoch*, *cookieB*, *challengeState*,  
*successorInstalled*, *revoked*, *oldLogoutHandled*⟩

*Next*  $\triangleq$   
 $\vee$  *StageTransfer*  
 $\vee$  *ConfirmTransfer*

$\vee \text{InstallCompetingSuccessor}$   
 $\vee \text{RunOldPageLogout}$   
 $\vee \text{RunOldPageCleanup}$

$\text{Spec} \triangleq \text{Init} \wedge \Box[\text{Next}]_{\text{vars}}$

$\text{TypeOK} \triangleq$   
 $\wedge \text{currentSession} \in \text{Sessions}$   
 $\wedge \text{epoch} \in 1 \dots 3$   
 $\wedge \text{cookieA} \in \text{Cookies}$   
 $\wedge \text{cookieB} \in \text{Cookies}$   
 $\wedge \text{challengeState} \in \text{ChallengeStates}$   
 $\wedge \text{successorInstalled} \in \text{BOOLEAN}$   
 $\wedge \text{revoked} \in \text{BOOLEAN}$   
 $\wedge \text{oldLogoutHandled} \in \text{BOOLEAN}$   
 $\wedge \text{oldCleanupHandled} \in \text{BOOLEAN}$

$\text{LiveSessionKeepsItsCookie} \triangleq$   
 $\wedge (\text{currentSession} = \text{"G1A"} \wedge \neg \text{revoked}) \Rightarrow \text{cookieA} = \text{"G1"}$   
 $\wedge \text{currentSession} = \text{"G2B"} \Rightarrow \text{cookieB} = \text{"G2"}$   
 $\wedge \text{currentSession} = \text{"G3A"} \Rightarrow \text{cookieA} = \text{"G3"}$

$\text{OldPageCannotRevokeSuccessor} \triangleq$   
 $\text{currentSession} \neq \text{"G1A"} \Rightarrow \neg \text{revoked}$

$\text{InstalledSuccessorCannotBeReplaced} \triangleq$   
 $\text{successorInstalled} \Rightarrow \text{currentSession} = \text{"G3A"}$

---